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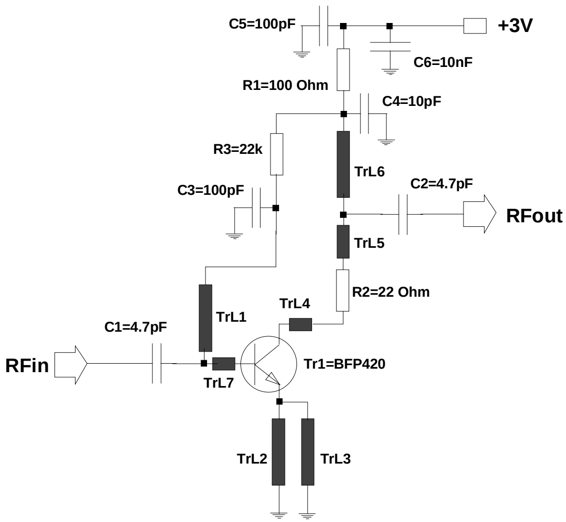
A Low-Noise-Amplifier at 900 MHz using SIEGET® BFP420

This application note describes a low noise amplifier at 900MHz using SIEMENS SIEGET®25 BFP420. The design emphasis has been on achieving a low noise figure. A circuit description, schematic, pcb layout and components list are shown below together with measured performance data.

Data at 0.9 GHz (3V and 5mA):
Gain=18 dB / IP3out=10dBm / NF=1.35dB / RLin-out >10dB



This amplifier at 900 MHz has been realized by using microstrip lines as matching elements. The design offers a good compromise between high IP3in- values, low noise figure and high return loss.

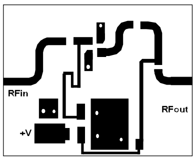
In order to optimise the design for a particular application please observe the following points:

- The layout size can be reduced by using chip inductors instead of the microstrip lines TrL1 and TrL6.
- Improved stabilization behaviour versus temperature and reduced variation in amplifier performance due to the device's Beta (current gain) distribution can be achieved by using an active bias circuit. Such a circuit is available as a single device from Siemens - BCR400. For further information please refer to Aplication Note 014. However the resistors R1 and R3 are sufficient in most applications for stabilization purposes.
- This circuit is not optimized for low noise figure, it is a first step to a good design. The measured figures include losses of SMA-connectors and the relatively high loss of the microstrip lines on the epoxy-board.The use of teflon material would provide an improvement of ≅ 0.1 dB.

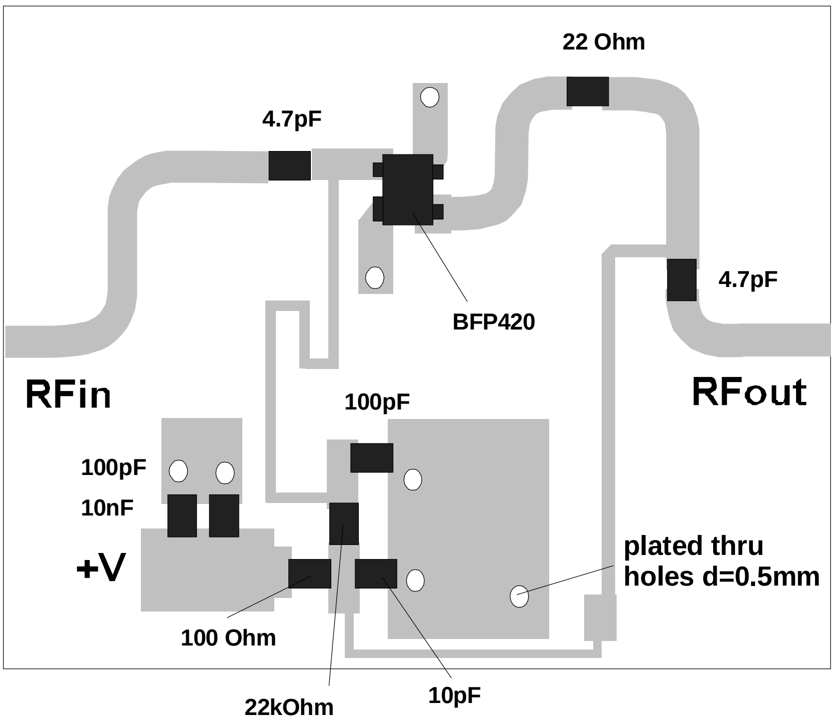
Resistor R2 is used to get higher rf-circuit-stability and return loss values at the output. It also affects the output intermodulation performance.

Component	Value	Unit	Size	Comment
R1	100	Ω	0603	bias / collector-resistance / VR1≅ 0.5V
R2	22	Ω	0603	to improve stability and output return loss
R3	22	kΩ	0603	bias / base-resistor
C1	4.7	pF	0603	inputmatch
C2	4.7	pF	0603	outputmatch
C3	100	pF	0603	rf-short
C4	10	pF	0603	outputmatch
C5	100	pF	0603	rf-short
C6	10	nF	0603	rf-short
Tr1			SOT343	SIEGET BFP420
TrL1				inputmatch and bias,w=0.3mm
TrL2				emitter-microstrip-line, w=0.95mm
TrL3				emitter-microstrip-line, w=0.95mm
TrL4				outputmatch, w=0.95mm
TrL5				outputmatch, w=0.95mm
TrL6				outputmatch and dc-bias, w=0.3mm
TrL7				inputmatch, w=0.95mm
Substrate	Fr4			h=0.5mm; εr=4.5

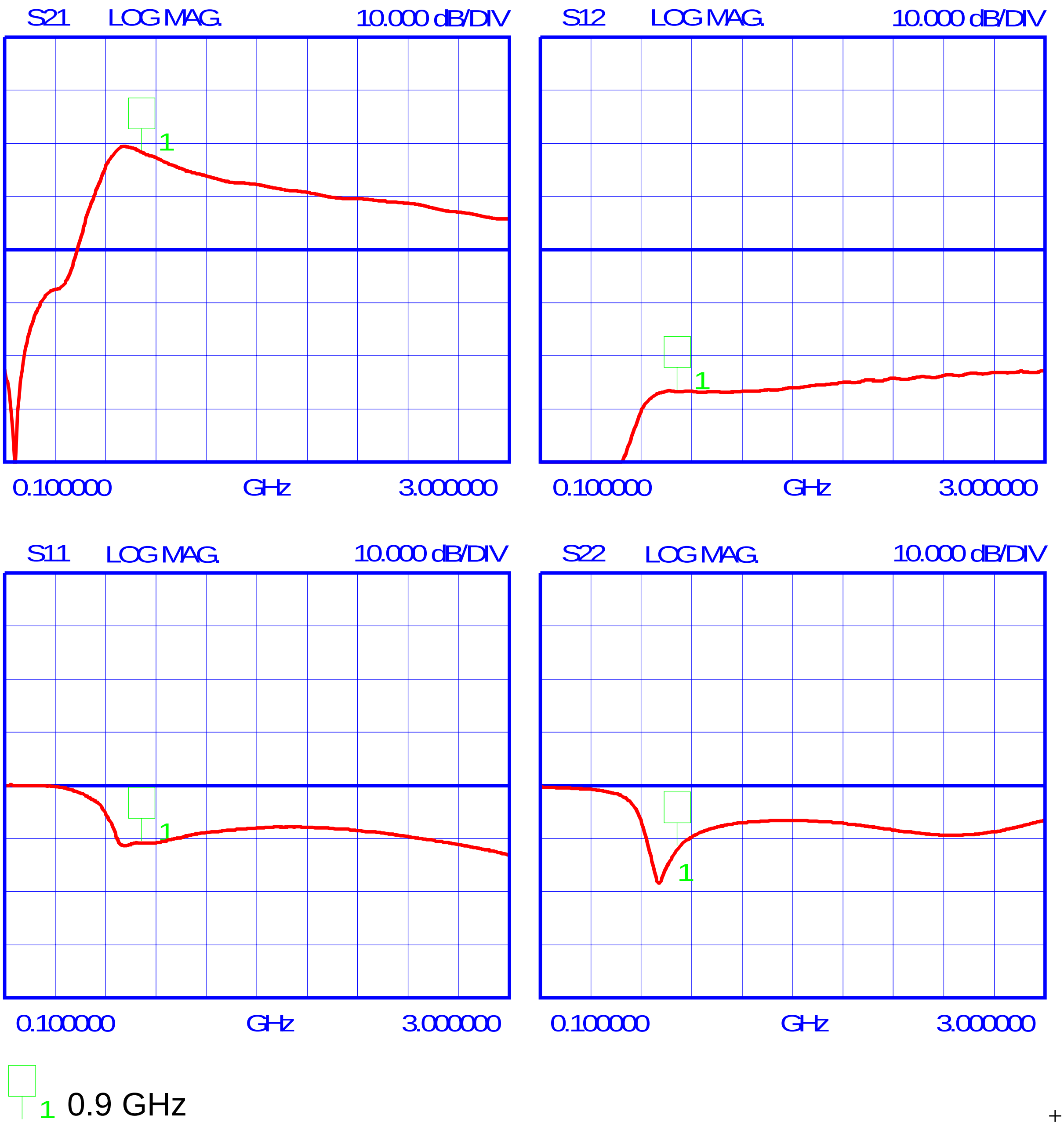
PCB Layout and Component Placement



scale 1:1
dim.: 25mm x 20mm



Measured data:



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